

SHAHZAIB SAQIB WARRAICH

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EDUCATION

University of Southern California, Viterbi School of Engineering

Los Angeles, CA

M.S., Applied Data Science - CGPA: 4.00

Jan 2024 - Present

Relevant Coursework: Machine Learning, Deep Learning, Applied Natural Language Processing, Database Management

EXPERIENCE

Graduate Research Assistant

Los Angeles, CA

Data, Interpretability, Language and Learning (DILL) lab

March 2024 - Present

Advisor: Dr. Swabha Swayamdipta

- Inferring practical implications of LLM data and model design choices by leveraging theoretical insights.
- Designing frameworks for LLM evaluation by building on a deeper theoretical understanding of language models.

Research Engineer II (ML/CV)

Redmond, WA

Retrocausal

Dec 2022 - Dec 2023

- Engineered precision assembly solutions with up to **98%** detection accuracy for industry leaders such as Schneider Electric, Carrier, and Honda using cutting-edge object detection and activity recognition algorithms.
- Led the development of an automated object detection model training pipeline, which employed the cut, paste, and mix augmentation technique to generate synthetic dataset alongside labeled data, thereby reducing manual data labeling time by almost **5 times**.

AI Engineer

San Jose, CA

Adlytic AI

June 2021 - Dec 2022

- Spearheaded extensive R&D on model optimization and pruning techniques, optimizing models through knowledge distillation and leveraging NVIDIA TensorRT to significantly reduce model inference latency at the edge by approximately **3 times**.
- Achieved high model throughput and performance in production, within **1.5%** of the original performance.
- Executed the development and deployment of a cloud-hosted recommender system pipeline, achieving a remarkable model accuracy benchmark of up to **97%**.

PROJECTS

Person Re-identification System | PyTorch, YOLOv5, FaceNet, Nvidia TensorRT, FastAPI

April 2022 - Dec 2022

- Leveraged facial recognition, super resolution, object detection, and multi-object tracking algorithms to design the model inference API. Utilized RTSP to perform object tracking on frames from live streams.
- Optimized object detection models to achieve an accuracy of up to **93%** with an IoU of nearly **0.77**.

Mobile App Recommender System | PySpark, Amazon SageMaker, EC2, Redis, Docker

June 2021 - Mar 2022

- Designed end-to-end system architecture on AWS (SageMaker, EMR, EC2, S3, SNS).
- Optimized matrix factorization models to generate collaborative and content based recommendations using Apache Spark.
- Stored and served recommendations using AWS managed Redis databases.
- Packaged and deployed application using Docker containers.
- Boosted the product's sale conversion rate by nearly **200%**.

PUBLICATIONS

- [Residual balanced attention network for real-time traffic Scene Semantic Segmentation](#), 2023, IJECE: Kherraki, SS Warraich, M Maqbool, R El Ouazzani

LEADERSHIP

Viterbi Graduate Student Association (VGSA)

Senator

Aug 2024 - Present

- Representing a student body of nearly 2,000 students, and planning and organizing professional development events.

NUST Model United Nations Society (NIMUN)

Director (Committees)

Nov 2020 - Apr 2021

- Led a 40-member team spanning liaison, operations, and logistics departments; effectively coordinated communication and efforts using Atlassian Jira, significantly increasing operational efficiency.

SKILLS & INTERESTS

Skills: Languages (Python, SQL), Deep Learning Frameworks (PyTorch, Hugging Face, vLLM), Machine Learning Frameworks (Scikit-Learn, OpenCV), Data Analysis Frameworks (Pandas, Dask, PySpark), Data Visualization (Matplotlib, Seaborn, Plotly), Cloud (AWS), Others (Docker, MongoDB, Git, LaTeX)

Interests: AI Ethics and Responsibility, Hackathons, User Experience (UX) Design